# PulseBoard: Project Status, Pending Tasks & Roadmap

Measuring Schedule, Effort & Flow for Small Agile Teams — CS447 Group 9 Evaluation Report

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| **Academic Course:** CS447 Software Project Management | **Repository:** github.com/S-creator31/Pulseboard |
| **Team Group 9:** Shlok Chauhan (202351131), Tushar Shukla (202351149), Keshav Badaya (202351065), Mohli Thapa (202351084) | **Active Git Branches:** main (up to date), feat/project-controls |
| **Delivered Scope:** 11 Production UI Surfaces, 6 Core Commitments Shipped | **Verification:** 6/6 Pure Unit Tests Passed \| Next.js Build 8/8 Routes Validated |

## 1. Executive Summary & Baseline Core Commitments (Delivered 100%)

PulseBoard transforms standard agile Kanban boards into an always-on **delivery management platform** by embedding classical project controls (Critical Path Method, Effort Tracking, Cumulative Flow, Cycle Time, and Fact-Grounded AI) without enterprise overhead. All **6 fixed core commitments** agreed upon in the project charter have been fully implemented, backed by a 90-day deterministic Mulberry32 dataset, pure algorithm unit tests, and production Next.js builds.

| Core Commitment (CS447 Charter) | Delivered Technical Architecture & UI Surface | Verification & Status |
|---|---|---|
| **1. Work Organization**<br>Board → List → Task Hierarchy | Full Kanban board with drag-and-drop, statusKey bidirectional synchronization (preventing status drift), density toggles (compact/cozy/comfortable), assignee & priority chips, and ■K command palette. | **100% SHIPPED**<br>**68 tasks, 5 lists, zero status drift.** |
| **2. Access & Audit Trail**<br>Role-Based Access & Audit Logging | 4-tier role hierarchy (Owner, Admin, Member, Viewer) with requireRole middleware; cursor-paginated Activity Drawer tracking card moves, time logs, and status transitions; rollback undo alerts. | **100% SHIPPED**<br>**186 historical transitions logged & auditable.** |
| **3. Schedule Visibility**<br>CPM Engine, Gantt & PERT Diagrams | Pure CPM scheduling engine with topological sort, forward/backward pass, and Total Float ($TF = LS - ES$). Interactive single-SVG Gantt timeline and ranked activity-on-node PERT diagram with Critical Path toggle. | **100% SHIPPED**<br>**6/6 unit tests passing (CPM math & cycle prevention).** |
| **4. Effort Visibility**<br>Logged Time vs Estimates & Capacity | Task estimate progress bars, quick-log popover, individual time audit ledger, and 4-week team capacity heatmap with 40-hour capacity baseline benchmarks and task hover breakdowns. | **100% SHIPPED**<br>**109 logged entries tracked against task estimates.** |
| **5. Flow Visibility**<br>Sprint Analytics & Bottlenecks | Four high-performance Mongoose aggregation pipelines: ideal vs actual sprint burndown, 4-sprint velocity trend with rolling average, cycle time scatter with p50/p85 percentiles, and cumulative flow diagram (CFD). | **100% SHIPPED**<br>**6 real-time delivery health status indicators.** |
| **6. AI-Assisted Insight**<br>Advisory & Fact-Grounded Insight | Fact-derived delivery digest (facts pre-aggregated server-side before prompt construction, metrics locked against hallucination) and Work Breakdown Assistant strictly bounded to 3–7 subtasks ≤8h each. | **100% SHIPPED**<br>**Deterministic offline fallback + schema unit tests.** |

## 2. Scope Trade-Offs & Architectural Decisions (Evaluator Review)

Under the 29-hour engineering budget, specific write-path complexities were deliberately structured as read-only computed views or deferred to Phase 2. This represents sound project control: prioritizing mathematical rigor and analytical visibility over complex UI mutations:

| Feature Item | Engineering Rationale & Trade-off Analysis | Academic Defense Line (Viva / Presentation) |
|---|---|---|
| **Drag-to-Reschedule on Timeline** (Interactive Bar Resizing on Gantt) | Direct canvas drag-rescheduling requires complex multi-task cascade recalculations in browser memory that risk invalidating topological sorting. Dates and durations are edited on cards/modals to preserve deterministic CPM invariants. | *'The timeline is an analytical projection over the scheduling engine; schedule modifications happen on task cards to preserve strict CPM DAG integrity.'* |
| **Dynamic Role Permission Editor** (Custom Matrix per User / Board) | A complete 4-tier RBAC system (Owner, Admin, Member, Viewer) is enforced via requireRole middleware. A dynamic matrix allowing custom permission bitmask editing was deferred in favor of fixed, battle-tested permission tiers. | *'Role permissions are strictly enforced at the API gateway layer; a dynamic permission rulebuilder is queued for Phase 2 multi-tenant administration.'* |
| **Secondary Write-Path Activity Logs** (List renames & metadata edits) | All core delivery events (card moves across lists, time logged, dependency links/unlinks) produce audit logs and status history entries. Cosmetic updates (list renaming) were deferred to prevent audit feed noise. | *'Audit logging captures all state transitions and effort entries; non-state cosmetic changes are batched in the database changelog.'* |

## 3. Engineering Backlog & High-Impact Improvements (Phase 2 Roadmap)

The following prioritized roadmap outlines key pending enhancements that build directly upon PulseBoard's core scheduling, flow, and analytics engines:

| Improvement Feature | Priority | Complexity | Target Impact & SPM Benefit |
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| **1. Real-time Multi-User Sync (Socket.io)** Live drag-and-drop card sync across concurrent users with active presence indicators. | **P1 (High)** | Medium (3d) | Eliminates polling; ensures sprint planning boards reflect immediate team-wide movements without race conditions. |
| **2. Three-Point PERT & Monte Carlo Risk** Allow entering optimistic (a), most likely (m), and pessimistic (b) durations with probability curves. | **P1 (High)** | Medium (2d) | Elevates deterministic CPM to probabilistic schedule forecasting ($E = \frac{a+4m+b}{6}$) with 95% confidence delivery dates. |
| **3. Automated Webhook & Slack Alerts** Push notifications for WIP cap breaches, sprint scope creep >15%, and critical path slippages. | **P2 (Med)** | Small (1d) | Provides proactive delivery alarms to project managers without requiring active dashboard polling. |
| **4. Executive PDF & CSV Report Generator** One-click downloadable delivery report containing sprint burndown, velocity tables, and audit logs. | **P3 (Norm)** | Small (1d) | Facilitates stakeholder status reporting, client milestone sign-offs, and academic grading documentation. |
| **5. Enterprise SSO & OAuth Providers** Sign in with Google Workspace, GitHub, or Microsoft Azure AD; organization multi-tenancy support. | **P3 (Norm)** | Medium (2d) | Streamlines enterprise adoption and academic lab onboarding with federated identity management. |